A3: Epidemic Spreading on Complex Networks

Complex Networks

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1 Introduction

Understanding how diseases, information, or computer viruses spread through interconnected systems is a central challenge in network science. The topology of a network dictates the dynamics of any spreading process operating upon it. In this project we explore these dynamics using the Susceptible-Infected-Susceptible (SIS) model, a standard compartmental framework used to study pathogens that do not confer long-lasting immunity.

This task has three main objectives. First, we aim to analyze how different network architectures influence epidemic outbreaks by comparing random Erdös-Rényi (ER) networks against scale-free Barabási-Albert (BA) networks across different average degrees and recovery probabilities. Second, we validate empirical Monte Carlo simulations against two theoretical frameworks: the Microscopic Markov Chain Approach (MMCA) and the Quenched Mean Field (QMF) model, evaluating their accuracy in predicting both the epidemic threshold and the stationary infection density. Finally, because large-scale stochastic simulations are highly computationally demanding, we address the practical challenges of network modeling. We propose an optimized algorithmic approach and compare the computational efficiency of interpreted execution (Python) versus compiled execution (C++) to achieve significant performance improvements.

2 The SIS model

The Susceptible-Infected-Susceptible (SIS) model is one of the simplest compartmental models for epidemic spreading. Each node in the network occupies one of two states at any given time:

- **Susceptible (S):** the node is healthy but can be infected.
- **Infected (I):** the node carries the disease and can transmit it to neighbors.

The dynamics are governed by two parameters:

1. $\beta \in [0, 1]$ is the transmission probability: at each discrete time step, an infected node transmits the disease to each susceptible neighbour independently with probability β.
2. $\mu \in [0, 1]$ is the recovery probability: at each time step, an infected node recovers (returns to the susceptible state) with probability μ. Because recovered nodes are susceptible again, the model admits a stationary endemic state in which a non-zero fraction of the population remains infected indefinitely.

The key macroscopic observable is the stationary infection density (the parameter that we will estimate in the experiments):

$$\rho = \frac{1}{N} \sum_{i=1}^{N} \rho_i, \tag{1}$$

where ρ_i is the long-run probability that node i is infected. An important characteristic of the SIS model is the existence of an epidemic threshold β_c. For $\beta < \beta_c$ the disease dies out ($\rho \to 0$), while for $\beta > \beta_c$ it persists at a positive endemic level.

2.1 Monte Carlo simulations

Monte Carlo (MC) simulations implement the SIS dynamics stochastically and average the resulting infection density over many independent realizations. The simulation is run for $T_{\max}$ time steps per realisation. The first T_{trans} steps are discarded as a transient, and ρ is estimated by averaging over the remaining $T_{\max} - T_{\text{trans}}$ steps. The procedure is repeated over N_{rep} independent realizations, each starting from a fresh random infection of a fraction R_0 of nodes, and the final estimate is the mean over all realizations.

Algorithm 1 shows the straightforward implementation. At each time step, the algorithm iterates over every susceptible node and checks, for each of its neighbors, whether that neighbour is infected and whether transmission occurs. The inner loop therefore touches every edge incident to a susceptible node, yielding a per-step cost of $O(N + E)$. With $N_{\text{rep}} \times T_{\max}$ steps across all realizations the total cost is $O(N_{\text{rep}} \cdot T_{\max} \cdot (N + E))$.

Algorithm 1 SIS Monte Carlo simulation (naive)

Require: Graph $G = (V, E)$, infection probability β, recovery probability μ, repetitions N_{rep}, initial fraction R_0, steps $T_{\max}$, transient T_{trans}
Ensure: Stationary infection density ρ

```
ρ_avg ← 0
for rep = 1 to N_rep do
    infected ← random sample of ⌊R_0 · N⌋ nodes
    susceptible ← V \ infected
    ρ_rep ← 0
    for t = 1 to T_max do
        if infected = ∅ then break
        end if
        recoveries ← ∅
        for v ∈ infected do
            if random() < μ then    recoveries ← recoveries ∪ {v}
            end if
        end for
        new_infected ← ∅
        for v ∈ susceptible do
            for u ∈ N(v) do
                if u ∈ infected and random() < β then
                    new_infected ← new_infected ∪ {v}
                    break
                end if
            end for
        end for
        if t > T_trans then
            ρ_rep ← ρ_rep + |infected|/N
        end if
        infected ← (infected \ recoveries) ∪ new_infected
        susceptible ← (susceptible \ new_infected) ∪ recoveries
    end for
    ρ_avg ← ρ_avg + ρ_rep/(T_max − T_trans)
end for
return ρ_avg/N_rep
```

The parameters used throughout this work are $N_{\text{rep}} = 100$, $R_0 = 0.2$, $T_{\max} = 1000$, and $T_{\text{trans}} = 900$, giving 100 post-transient steps per realisation.

2.2 Microscopic Markov Chain Approach (MMCA)

The MMCA [Gómez et al., 2010] provides a deterministic approximation to the SIS dynamics by tracking, for each node i, the probability $\rho_i^{(t)}$ of being infected at time t. The update equation is obtained by writing the

Markov transitions exactly:

$$\rho_i^{(t+1)} = \rho_i^{(t)}(1-\mu) + \left(1-\rho_i^{(t)}\right)\left[1 - \prod_{j\in\mathcal{N}(i)}\left(1-\beta\rho_j^{(t)}\right)\right]. \tag{2}$$

The first term on the right accounts for infected nodes that do not recover. The second term accounts for susceptible nodes $(1-\rho_i^{(t)})$ that are infected by at least one neighbour. The key assumption is that the states of different nodes are statistically independent, which allows the joint probability of escaping infection from all neighbors to be factored as a product. Unlike mean-field approaches that linearise this product, MMCA retains the full product form, making it exact on trees and a good approximation on locally tree-like networks.

Starting from $\rho_i^{(0)} = 0.5$ for all i, the iteration is run until

$$\max_i \left|\rho_i^{(t+1)} - \rho_i^{(t)}\right| < \varepsilon, \tag{3}$$

with $\varepsilon = 10^{-10}$, and the global density $\rho = \frac{1}{N}\sum_i \rho_i$ is recorded.

For computational efficiency the product over neighbors is evaluated using the log-sum trick:

$$\prod_{j\in\mathcal{N}(i)}(1-\beta\rho_j) = \exp\left(\sum_j A_{ij}\log(1-\beta\rho_j)\right) = \exp\big(A\cdot\log(1-\beta\rho)\big)_i, \tag{4}$$

reducing the per-iteration cost to a single sparse matrix–vector multiplication.

Linearizing the update equation around $\rho = 0$ shows that the endemic fixed point bifurcates when the spectral radius of βA equals μ, yielding the epidemic threshold

$$\beta_c^{\mathrm{MMCA}} = \frac{\mu}{\lambda_{\max}(A)}. \tag{5}$$

2.3 Quenched Mean Field (QMF)

The QMF approach [Wang et al., 2003, Chakrabarti et al., 2008] (often referred to as the Nonlinear Dynamical System (NLDS) approach in the literature) also retains the full adjacency matrix but replaces the product infection probability with a linear mean-field approximation. Writing the continuous-time rate equation:

$$\frac{d\rho_i}{dt} = -\mu\rho_i + \beta(1-\rho_i)\sum_j A_{ij}\rho_j, \tag{6}$$

the term $\beta\sum_j A_{ij}\rho_j$ acts as the effective force of infection on node i. Setting $d\rho_i/dt = 0$ and defining $\theta_i = \sum_j A_{ij}\rho_j = (A\rho)_i$ yields the self-consistent fixed-point equation:

$$\rho_i = \frac{\beta\theta_i}{\mu+\beta\theta_i}. \tag{7}$$

This is solved by the iteration: initialise $\rho^{(0)} = 0.5$, then alternately compute $\theta = A\rho$ and update $\rho_i \leftarrow \beta\theta_i/(\mu+\beta\theta_i)$ until convergence. Each iteration again requires only a single matrix–vector multiplication.

The epidemic threshold is identical to that of MMCA:

$$\beta_c^{\mathrm{QMF}} = \frac{\mu}{\lambda_{\max}(A)}. \tag{8}$$

Because QMF linearizes the infection probability, it overestimates ρ relative to MMCA for β well above threshold. Both methods share the same threshold β_c, so they agree in the critical region but diverge at higher infection rates.

3 Networks used

All experiments are conducted on four networks of $N = 1000$ nodes generated with fixed random seeds for reproducibility.

Erdös–Rényi (ER). In an ER graph $G(N, p)$ each pair of nodes is connected independently with probability p. Choosing $p = \langle k \rangle/(N-1)$ targets a given average degree $\langle k \rangle$. The degree distribution follows a Poisson distribution centred on $\langle k \rangle$. The spectral radius satisfies $\lambda_{\max} \approx \langle k \rangle$, and the epidemic threshold scales as $\beta_c \approx \mu/\langle k \rangle$ (Equation 8 in [Gómez et al., 2010])

Barabási–Albert (BA). BA graphs are grown by preferential attachment. Each new node attaches m edges to existing nodes with probability proportional to their current degree, producing a power-law degree distribution $P(k) \sim k^{-3}$. β_c is lower compared to ER graphs with the same average degree.

Two values of average degree are studied for each model: $\langle k \rangle \approx 4$ and $\langle k \rangle \approx 6$. The detailed statistics of each network are reported in Table 1.

Network	N	E	$\langle k \rangle$	$\lambda_{\max}$	β_c (μ=0.2)	β_c (μ=0.4)
ER $\langle k \rangle = 4$	1000	1988	3.98	5.213	0.0384	0.0767
ER $\langle k \rangle = 6$	1000	2991	5.98	7.178	0.0279	0.0557
BA $\langle k \rangle = 4$	1000	1996	3.99	10.823	0.0185	0.0370
BA $\langle k \rangle = 6$	1000	2991	5.98	14.237	0.0140	0.0281

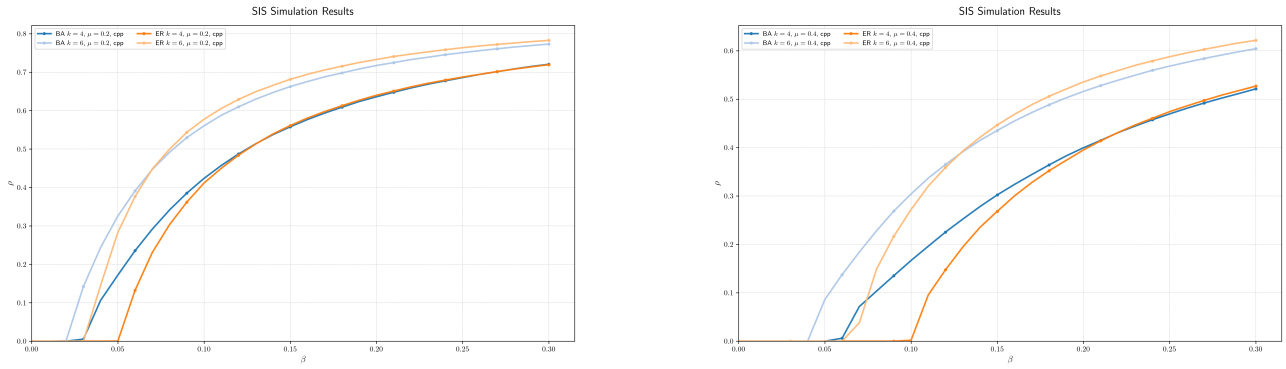
Table 1: Detailed statistics of the four networks used in the experiments. $\langle k \rangle$ is average degree. $\lambda_{\max}$ is the spectral radius of the adjacency matrix. $\beta_c = \mu/\lambda_{\max}$ is the epidemic threshold.

Two key observations can be derived from Table 1. First, for the same $\langle k \rangle$, BA networks have a much larger $\langle k^2 \rangle$ and $\lambda_{\max}$ than ER networks, reflecting the presence of high-degree hubs. This translates directly into a lower β_c: BA networks are considerably more vulnerable to epidemic spreading than their ER counterparts. Second, increasing $\langle k \rangle$ from 4 to 6 increases $\lambda_{\max}$ in both models, again lowering the threshold.

4 Results and Discussion

In this section we evaluate the results of our epidemic spreading models. We first explore how the underlying network topologies and parameters affect the spread of a disease using stochastic Monte Carlo simulations. Then, we assess the accuracy of theoretical models by comparing their predictions to our empirical data.

4.1 Epidemic spreading: ER vs. BA networks



(a) Recovery probability $\mu = 0.2$

(b) Recovery probability $\mu = 0.4$

Figure 1: Comparison of the stationary infection density ρ as a function of the transmission probability β for Erdös-Rényi (ER) and Barabási-Albert (BA) networks. Each plot illustrates the epidemic spreading dynamics across networks with average degrees $\langle k \rangle = 4$ and $\langle k \rangle = 6$, generated via Monte Carlo simulations.

Figure 1 shows the stationary infection density ρ as a function of the transmission probability β, comparing the Erdös-Rényi (ER) and Barabási-Albert (BA) networks for two different aveage degrees $\langle k \rangle \in \{4, 6\}$ and two recovery probabilities $\mu \in \{0.2, 0.4\}$.

The main difference between the two network topologies is the start of the epidemic outbreak. For any value of $\langle k \rangle$ and μ, the transmission probability β needed to trigger an epidemic is much smaller in BA than in ER networks. In particular, looking at Figure 1 ($\mu = 0.2$), the BA network with $\langle k \rangle = 6$ begins to show a non-zero infection density at $\beta \approx 0.014$, while the corresponding ER network does so later, at $\beta \approx 0.020$. This perfectly aligns with the theoretical predictions for the epidemic threshold (β_c) calculated from the spectral radius in Table 1.

This difference is mainly due to the degree distributions of each of the networks. On the one hand, ER networks have a Poisson degree distribution, meaning that most nodes have a degree very similar to the average $\langle k \rangle$. On the other hand, BA networks are generated through preferential attachment, which results in a power-law distribution characterized by the presence of hubs. These high-degree nodes are the main responsibles for the fast spread of the disease. Given their high degree, they are more likely to become infected early in the dynamics, and to quickly transmit the disease to a large number of neighbours. Therefore, BA networks are more vulnerable to epidemic spreading, allowing even diseases with low infectivity to survive.

Within the same network model (ER or BA), increasing the average degree from $\langle k \rangle = 4$ to $\langle k \rangle = 6$, produces a shift the epidemic curve to the left. In a denser network, individuals have more contacts, which provides the disease with more opportunities to spread.. As a result, a lower critical transmission probability (β_c) is needed to trigger an epidemic, leading to a higher stationary infection density (ρ) for any given value of β.

Regarding the effect of the recovery probability, we compare Figure 1a ($\mu = 0.2$) and Figure 1b ($\mu = 0.4$). When μ is doubled, it means that individuals are able to recover from the disease twice as fast. To compensate for the shorter infectious period and to maintain a stationary endemic state, the transmission probability β must also double. This shifts all the epidemic curves to the right, delaying the onset of the epidemic but still maintaining the structural differences between ER and BA topologies.

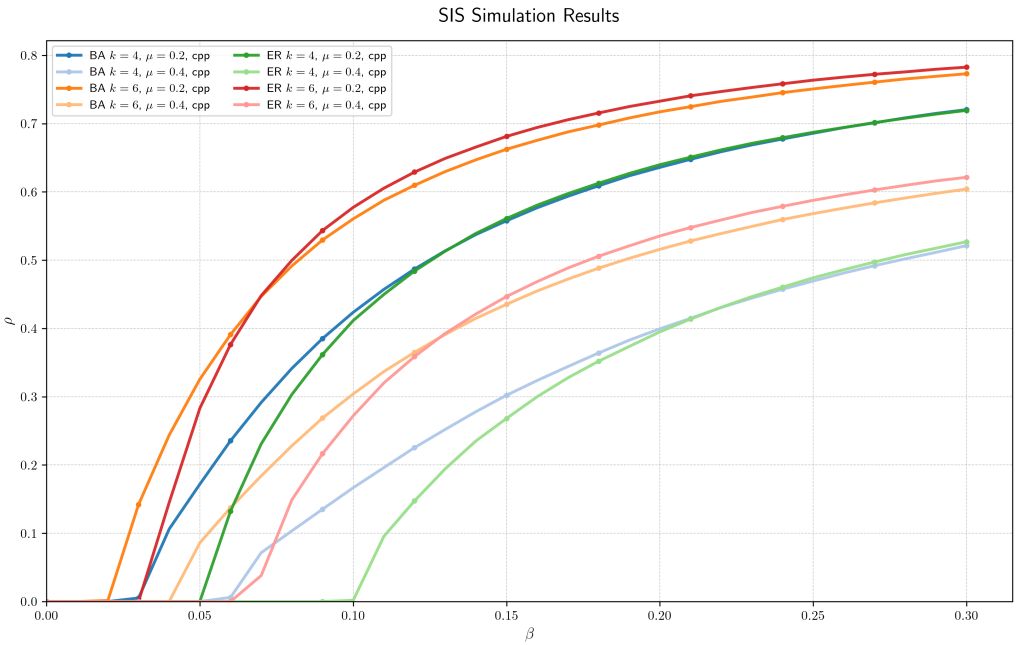


Figure 2: Global comparison of epidemic spreading dynamics for all eight experimental combinations, considering network topology (ER vs. BA), average degree ($\langle k \rangle \in \{4, 6\}$), and recovery probability ($\mu \in \{0.2, 0.4\}$).

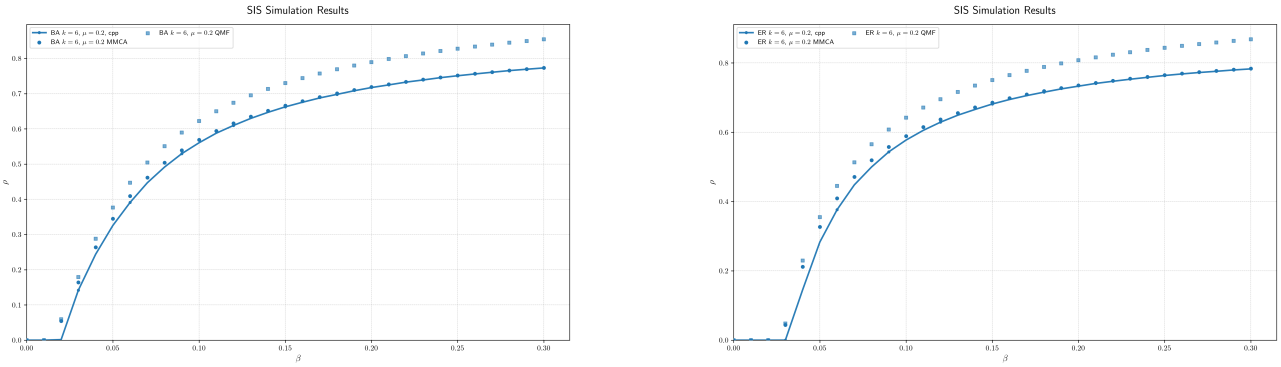
Finally, Figure 2 provides a visual summary by overlaying all eight experimental combinations on a single plot. This combined plot makes it easy to compare the overall risk. By looking at the order of the curves from left to right, we can clearly tell which setup is the most vulnerable to an epidemic and which is the most resistant. The network most vulnerable to an epidemic is the BA network with a higher average degree and a lower recovery probability (BA, $\langle k \rangle = 6$, $\mu = 0.2$), while the most robust is the ER network with a lower average degree and a

higher recovery probability (ER, $\langle k \rangle = 4$, $\mu = 0.4$).

Moreover, the plot highlights the relative importance of each variable. While doubling the recovery rate (μ) delays the outbreak, switching the network structure from ER to BA has a much larger effect on the epidemic threshold. This demonstrates that the underlying network topology is the primary factor for disease vulnerability.

Finally, although the two topologies behave very differently at the start of an outbreak, they converge at high infection rates. As the transmission probability (β) approaches 0.30 (from ≈ 0.15 onwards), the curves for the ER and BA networks overlap almost entirely. This indicates that when a disease is highly contagious, the overall infection density saturates, making the specific network structure less relevant.

4.2 Theoretical models vs. Monte Carlo simulations



(a) BA network ($\langle k \rangle = 6$, $\mu = 0.2$)

(b) ER network ($\langle k \rangle = 6$, $\mu = 0.2$)

Figure 3: Comparison of the stationary infection density ρ obtained from Monte Carlo simulations (C++) against theoretical predictions from the MMCA and QMF models. The plots show the dynamics as a function of the transmission probability β for a fixed recovery probability $\mu = 0.2$.

In order to validate the empirical results of our epidemic spread model, we compare the infection density ρ obtained from the Monte Carlo simluations against the theoretical predictions of the MMCA and QMF models. Figure 3 shows the comparison for BA and ER networks with fixed average degree of $\langle k \rangle = 6$ and also fixed recovery probability $\mu = 0.2$. We focus on this particular setup to provide a clear and direct comparison. Modifying the network hyperparameters ($\langle k \rangle$ or μ) shifts the exact placement of the epidemic curves, but it does not alter the behavior or the relative accuracy of the theoretical models discussed in this section.

The first thing we can notice is the perfect alignment of the epidemic threshold. As stated in Sections 2.2 and 2.3, both theoretical models (MMCA and QMF) calculate the critical onset point of an epidemic in the same way ($\beta_c = \mu/\lambda_{max}$). In both subfigures, the Monte Carlo simulation transitions from a healthy, disease-free state to an infected state ($\rho > 0$) exactly at this predicted theoretical value, which confirms the accuracy of the theoretical calculations for both types of networks.

Although the QMF model accurately predicts the starting point of the epidemic, it overestimates the infection density (ρ) as the transmission probability (β) increases. We can see this gap get wider in both networks at higher values of β. This overestimation happens because the QMF model simplifies the mathematical dynamics. In particular, it assumes that the infection states of neighboring nodes are completely independent. In a real scenario, if a disease is widespread, an infected node's neighbors are also likely to be sick. By ignoring these local clusters (dynamical correlations), the QMF model assumes there are always plenty of susceptible neighbors to infect, which inflates the predicted size of the epidemic.

In contrast, the MMCA model matches the Monte Carlo results very closely across the whole range of β. This high accuracy comes from how the model is mathematically structured. Instead of using a linear approximation (like QMF does), MMCA calculates the exact probability that a healthy node avoids getting infected from each of its neighbours. Since probabilities are evaluated for each individual node and without any mathematical shortcuts, MMCA is able to capture the spreading dynamics much more realistically. We can say that it serves as an accurate theoretical upper bound for real-world simulations.

5 Code Efficiency

The naive Monte Carlo implementation (Algorithm 1) has two main bottlenecks. First, infected and susceptible nodes are stored as Python lists, making membership queries $O(N)$. Second, the inner loop iterates over all susceptible nodes at every time step, even though most of them have no infected neighbour and therefore zero probability of becoming infected.

Algorithm 2 shows the optimized implementation that addresses both issues. The optimized algorithm was first written in Python and then translated to C++ without any algorithmic changes [1].

Algorithm 2 SIS Monte Carlo simulation (optimized)

Require: Graph $G = (V, E)$, parameters β, μ, N_{rep}, R_0, $T_{\max}$, T_{trans}
Ensure: Stationary infection density ρ

1: Pre-compute $\mathcal{N}(v)$ as a **set** for all $v \in V$ ▷ done once before the rep loop
2: $\rho_{\text{avg}} \leftarrow 0$
3: **for** rep $= 1$ **to** N_{rep} **do**
4: infected $\leftarrow$ random sample of $\lfloor R_0 \cdot N \rfloor$ nodes, stored as a **hash-set**
5: $\rho_{\text{rep}} \leftarrow 0$
6: **for** $t = 1$ **to** $T_{\max}$ **do**
7: **if** infected $= \emptyset$ **then break**
8: **end if**
9: recoveries $\leftarrow \{v \in \text{infected} : \text{random}() < \mu\}$
10: candidates $\leftarrow \left(\bigcup_{v \in \text{infected}} \mathcal{N}(v)\right) \setminus \text{infected}$ ▷ only susceptible neighbors of infected nodes
11: new_infected $\leftarrow \emptyset$
12: **for** $v \in$ candidates **do**
13: $m \leftarrow |\mathcal{N}(v) \cap \text{infected}|$ ▷ number of infected neighbors
14: **if** random() $< 1 - (1 - \beta)^m$ **then**
15: new_infected $\leftarrow$ new_infected $\cup \{v\}$
16: **end if**
17: **end for**
18: **if** $t \geq T_{\text{trans}}$ **then**
19: $\rho_{\text{rep}} \leftarrow \rho_{\text{rep}} + |\text{infected}|/N$
20: **end if**
21: infected $\leftarrow$ (infected $\setminus$ recoveries) $\cup$ new_infected
22: **end for**
23: $\rho_{\text{avg}} \leftarrow \rho_{\text{avg}} + \rho_{\text{rep}}/(T_{\max} - T_{\text{trans}})$
24: **end for**
25: **return** $\rho_{\text{avg}}/N_{\text{rep}}$

The optimized version introduces four key changes with respect to the naive one:

1. **Hash-set storage.** The infected set is maintained as a hash-set (Python `set` / C++ `unordered_set`), reducing membership checks from $O(N)$ to $O(1)$.

2. **Candidate filtering.** Instead of iterating over all susceptible nodes, only the union of the neighbourhoods of currently infected nodes is considered (line 9). When the epidemic is sparse, this set is much smaller than N, providing a large practical speedup.

3. **Compound infection probability.** Rather than drawing a separate Bernoulli trial for each infected neighbour, the probability of being infected by at least one of m infected neighbors is computed in closed form as $1 - (1 - \beta)^m$ (line 12), reducing the number of random number draws.

4. **List comprehensions (Python only).** The recoveries and candidate sets are built with Python set/list comprehensions instead of explicit `for` loops. Comprehensions execute in CPython's internal C loop, which is faster than interpreted Python loops.

[1] Given that this was not the main scope of the project, the translation to C++ was carried out with the use of LLMs.

5.1 Timing comparison

Table 2 reports the total wall-clock time to run all 248 simulation points (4 networks $\times$ 2 values of μ $\times$ 31 values of β) with $N_{\text{rep}} = 100$ on the same hardware [2]. Three implementations are compared: the naive Python baseline, the optimized Python version applying the three changes described above, and the C++ port of that same optimized algorithm (identical logic, no further algorithmic changes).

Implementation	Total time	Speedup vs. naive	Speedup vs. efficient Py
Naive Python	$\approx$53544 s ($\sim$14.9 h)	1$\times$	—
Efficient Python	$\approx$4606 s ($\sim$1.3 h)	11.6$\times$	1$\times$
Efficient C++	$\approx$1162 s ($\sim$19 min)	46.1$\times$	$\sim$4$\times$

Table 2: Total wall-clock time for the full simulation campaign (248 points, $N_{\text{rep}} = 100$ each). Speedup is relative to the naive Python baseline.

The optimized Python implementation is roughly 11.6$\times$ faster than the naive version. The C++ implementation applies the same algorithmic optimisations as the efficient Python version, but takes advantage from compiled execution, native data structures, and the absence of interpreter overhead. This yields an additional $\sim 4\times$ speedup over efficient Python, bringing the total reduction to $\sim$46$\times$ relative to the naive baseline. The C++ run completes all the experiments in under 20 minutes, compared to nearly 15 hours for the naive implementation.

5.2 Implementation correctness

Despite the algorithmic differences described above, the three implementations are equivalent. The slight discrepancies visible in Figure 4 are purely due to Monte Carlo sampling variance. Each implementation consumes random numbers at a different rate per time step (the naive version draws up to one number per infected neighbour per susceptible node and the optimized version draws exactly one per candidate), so despite sharing the same initial seed the two random streams diverge immediately, producing different stochastic trajectories. The discrepancy is most visible at large β, where the endemic state fluctuates more across realizations, amplifying the error.

6 Conclusion

This project shows the impact that network topology has on epidemic dynamics, while highlighting the computational requirements of simulating such complex systems.

From a structural perspective, our results confirm that Barabási-Albert networks are more vulnerable to epidemic spreading than Erdös-Rényi networks. The preferential attachment mechanism of BA networks generates a power-law degree distribution with extreme hubs. These highly connected nodes act as the big spreaders of the infection, lowering the epidemic threshold (β_c) and allowing even weak pathogens to survive. Moreover, increasing the network density ($\langle k \rangle$) across both topologies provides additional transmission pathways, facilitating the spread of the disease. In contrast, a higher recovery probability (μ) proportionally increases the transmission rate required to maintain the infection.

Regarding the theoretical models, both MMCA and QMF accurately predict the critical epidemic threshold. However, QMF overestimates the total outbreak size because it oversimplifies the mathematical interactions between neighboring nodes. In contrast, MMCA evaluates these infection probabilities exactly, allowing it to capture the local spreading dynamics much more accurately. As a result, MMCA closely matches the empirical Monte Carlo simulations and provides a reliable theoretical upper bound for both network topologies.

Finally, while computational efficiency was not the primary focus of this project, our implementation highlights its practical importance. By optimizing the simulation algorithm and porting it to a compiled language (C++), we achieved a $\sim 46\times$ speedup. This reduced the total runtime from nearly 15 hours to under 20 minutes, demonstrating that efficient code is essential for making large-scale stochastic modeling feasible

[2] All simulations were run on a Linux machine, running Ubuntu with an AMD Ryzen 9 7900X 12-core processor

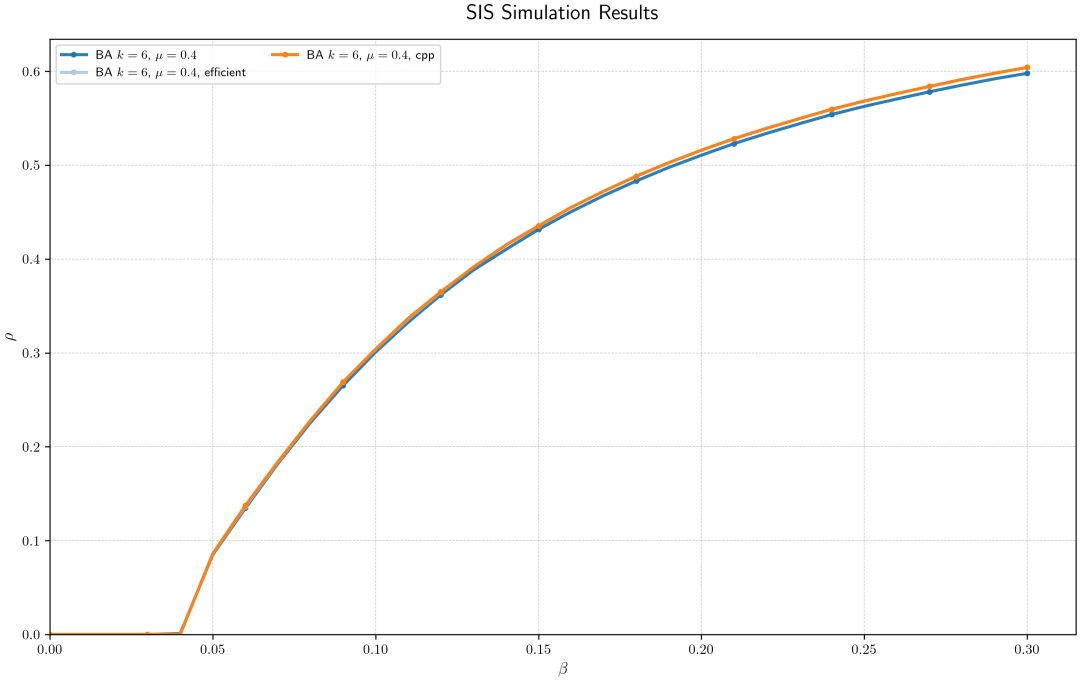


Figure 4: Stationary infection density ρ as a function of β for the BA network with $\langle k \rangle = 6$ and $\mu = 0.4$, estimated by the three implementations. The curves overlap almost perfectly. Differences are due to Monte Carlo sampling variance from finite $N_{\text{rep}} = 100$.

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